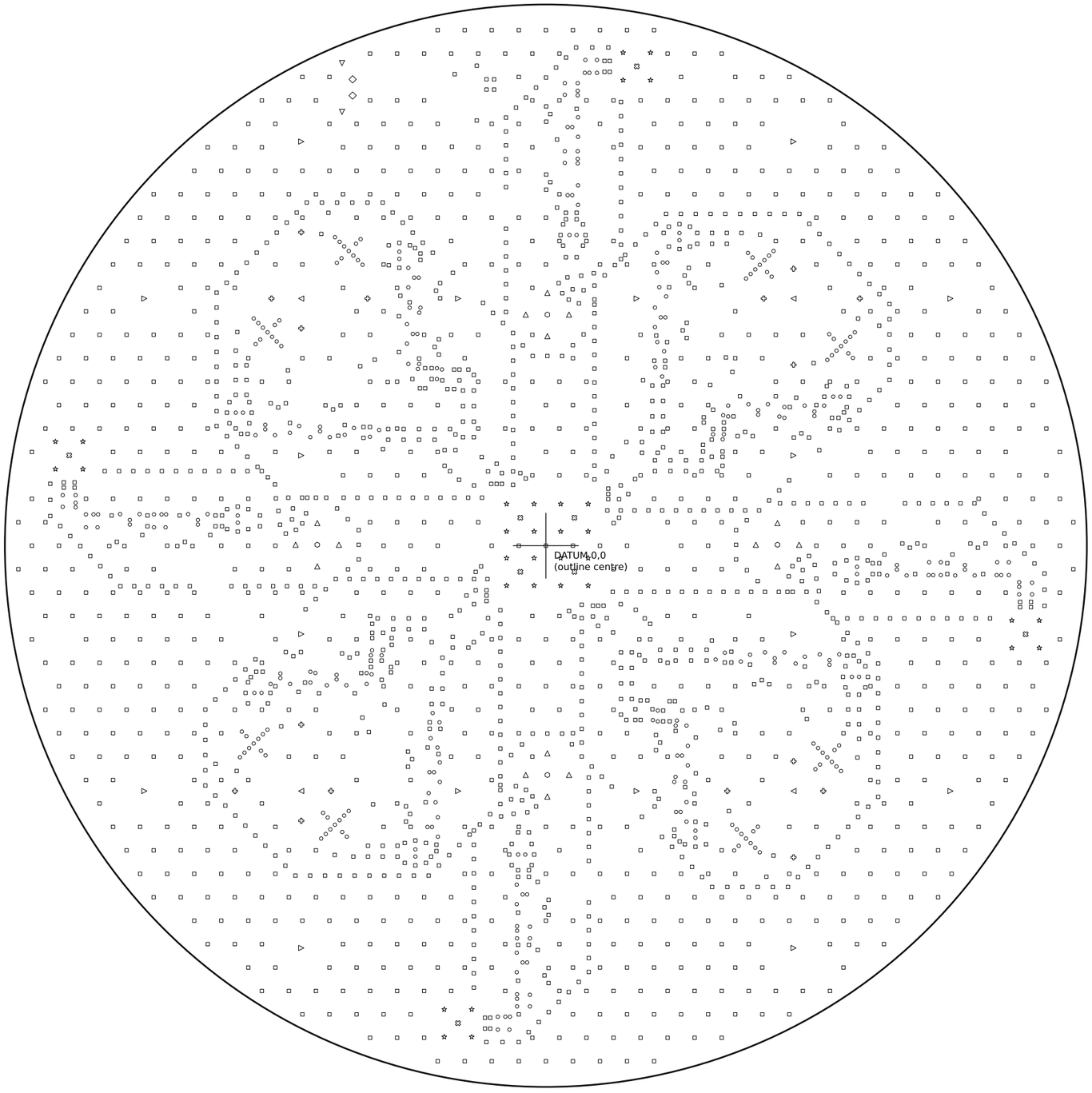
Holes shown with the drill symbol of the table; via fence and stitching vias D0.30 / D0.25.



DRILL TABLE (finished hole, mm)

SYM	DRILL	TYPE	QTY	NOTE
o	0.25	PTH	328	vias
s	0.30	PTH	1853	vias
^	0.77	PTH	16	
D	1.02	PTH	2	
v	1.27	NPTH	2	
P	1.35	PTH	16	
X	1.50	PTH	8	
*	1.70	PTH	32	
h	2.20	PTH	4	
<	3.20	PTH	4	
>	3.60	PTH	16	

STACK-UP (1.6 mm, order exactly this)

LAYER	TYPE	mm	er	NOTE
F.Mask	mask	0.010	3.8	green LPI
F.Cu	copper	0.035		L1: solid GND = antenna ground plane
Core1	core	0.710	4.4	FR-4 high-Tg core (IT-180A)
In1.Cu	copper	0.035		L2: GND + per-channel +5V star traces
PP1	prepreg	0.540	4.3	FR-4 prepreg 2x2116 + 1x1080 (bonds FR-4 to
In2.Cu	copper	0.035		L3: solid GND, RF reference for L4
Core2	core	0.168	3.66	Rogers RO4350B 0.168 mm (6.6 mil), tan d 0.0
B.Cu	copper	0.035		L4: RF lines + all SMD parts, ENIG
B.Mask	mask	0.010	3.8	green LPI

NOTES

1. IPC-6012 / IPC-A-600 Class 2 (Class 3 if airborne). UL 94V-0. RoHS.
2. FR-4 high Tg >= 170 C (IT-180A / S1000-2) + Rogers RO4350B 0.168 mm core, 1 oz. Plasma desmear.
3. 50 ohm +-10 % on L4: 0.34 mm track over L3 (RO4350B), with mask. TDR coupon on the panel.
4. ENIG. Green LPI mask, white legend both sides. No mask on the shield-frame GND lands (gasket).
5. Vias in pads (D0.25 / 0.50) resin filled and capped, IPC-4761 Type VII. Annular ring >= 0.125.
6. Track / space >= 0.15 / 0.15 mm. Copper to board edge >= 0.5 mm.
7. Bow and twist <= 0.5 %. 100 % electrical test against CRL-LS-CRPA4.d356.
8. Data: CRL-LS-CRPA4_Gerbers.zip (Gerber X2 + Excellon), drill map, IPC-D-356.

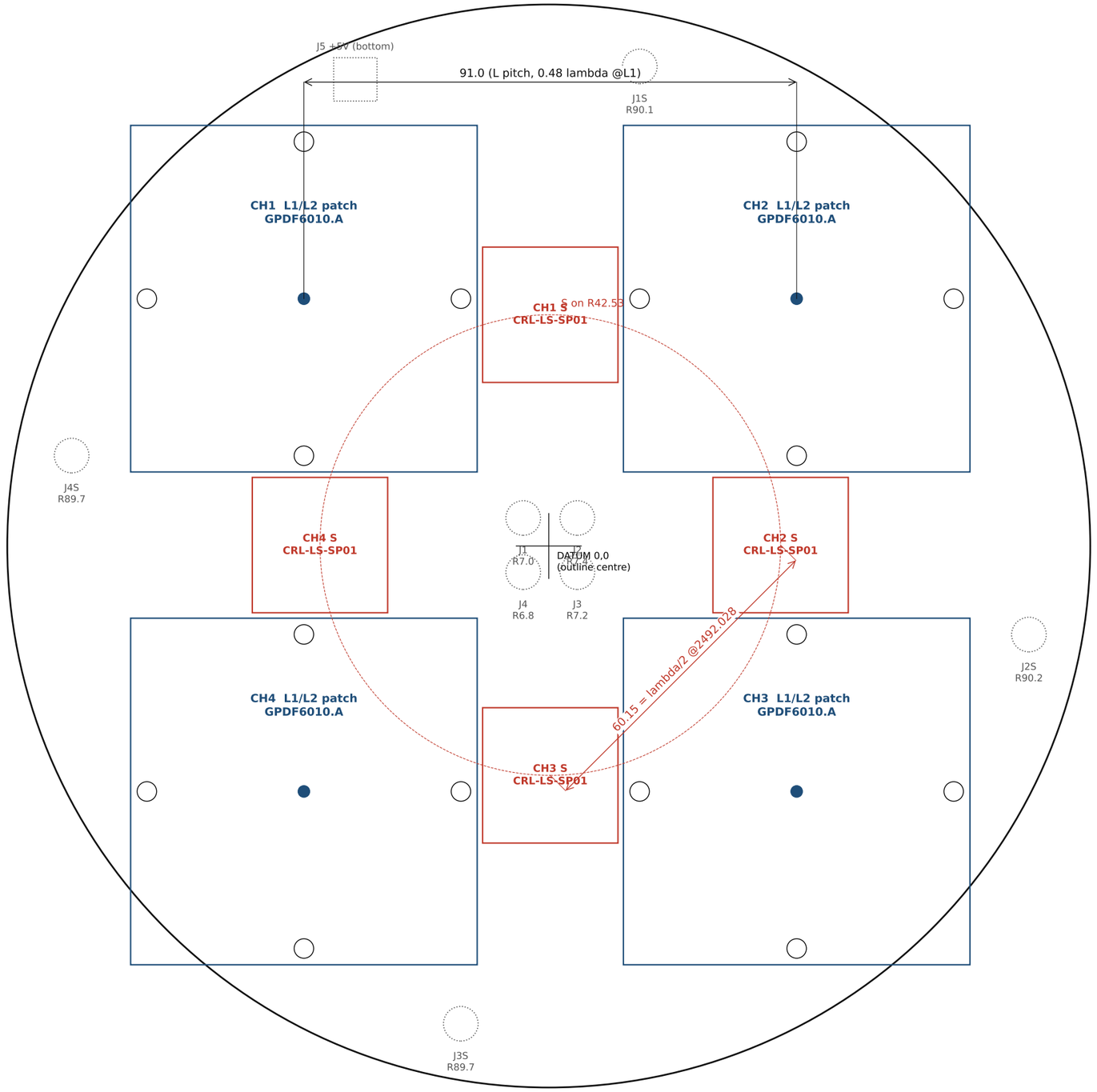
BEL - CRL

ANTENNA / RFM LAB

L1 + L2/L5 + NavIC S 4-ch CRPA front end

PCB fabrication drawing (top view)

PART	DWG	REV	SHEET
CRL-LS-CRPA4 PCB	CRL-LS-CRPA4-DWG	A	1 / 6
DATE	UNITS	PROJ.	TOL.
06-11-2026	mm	1st angle	ISO 2768-mK



POSITIONS (frame above)

ITEM	X	Y	R array
CH1 L patch	-45.22	+45.67	64.35
CH2 L patch	+45.78	+45.67	64.35
CH3 L patch	+45.78	-45.33	64.35
CH4 L patch	-45.22	-45.33	64.35
CH1 S patch	+0.28	+42.70	42.53
CH2 S patch	+42.81	+0.17	42.53
CH3 S patch	+0.28	-42.36	42.53
CH4 S patch	-42.25	+0.17	42.53
J1 SMA	-4.72	+5.17	-
J1S SMA	+16.81	+88.56	-
J2 SMA	+5.28	+5.17	-
J2S SMA	+88.67	-16.36	-
J3 SMA	+5.28	-4.83	-
J3S SMA	-16.25	-88.22	-
J4 SMA	-4.72	-4.83	-
J4S SMA	-88.11	+16.70	-
J5 DC	-35.72	+86.17	-

NOTES

1. Array centre (+0.280, +0.170) from the customer board combine_model_2.0.
2. Sequential rotation: element k is element 0 turned -90 deg x k about the array centre.
3. L patches: 4 x M3 through D3.6 PTH GND at +-29 mm; centre screw D3.2 grounded.
4. S patches CRL-LS-SP01 (BEL CRL): <= 25 x 25 mm, 4 pins D0.77 on R4,
   centre D2.2 GND; feeds F0 (+x, N element) -> hybrid pin 4, F90 (-y) -> pin 6 = RHCP.
5. Dotted: parts on the bottom side (SMA, DC) shown for reference.
6. 8-element growth path: see the project notes (D260-280 mm board or smaller patches).

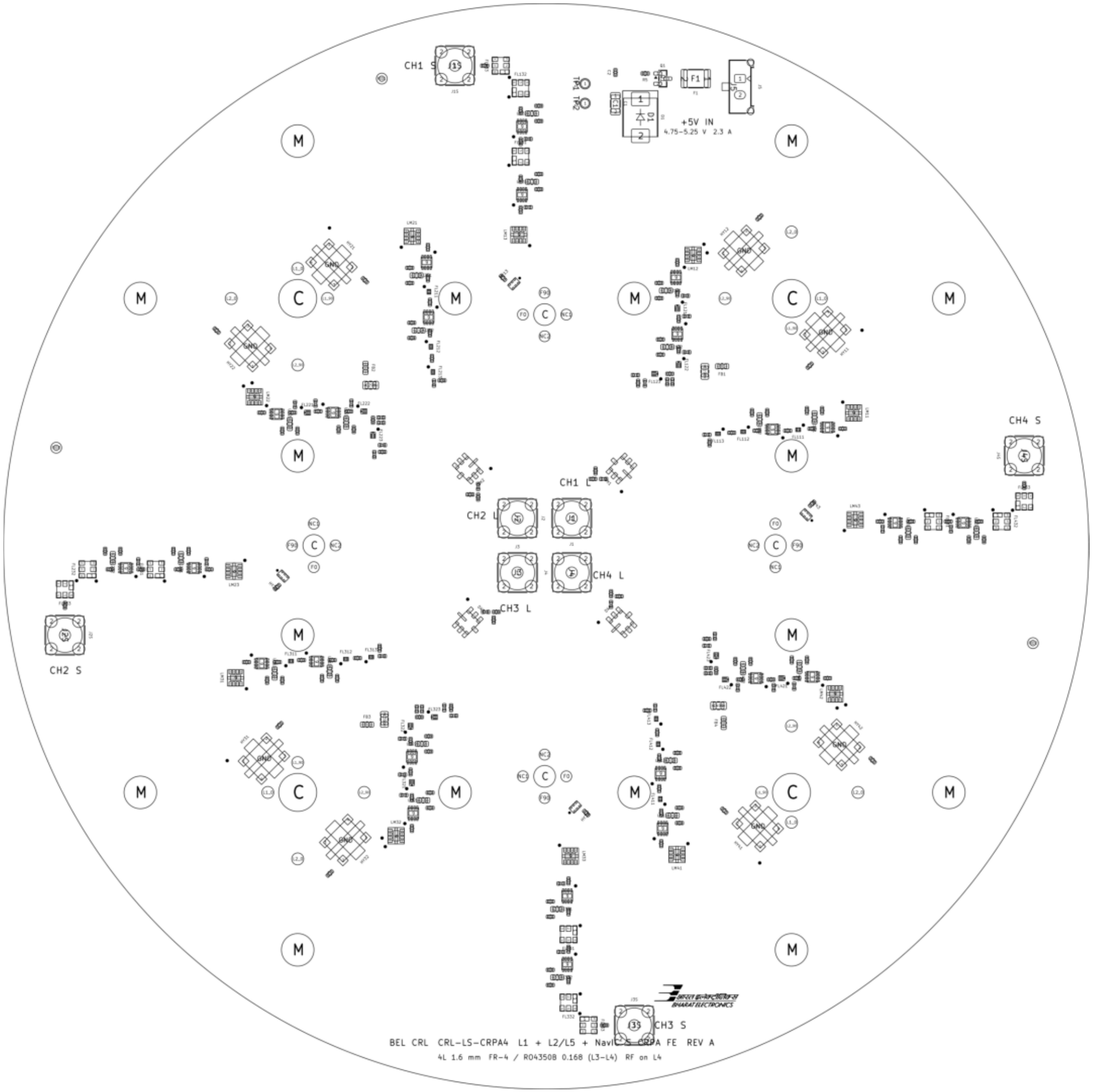
BEL - CRL

ANTENNA / RFM LAB

L1 + L2/L5 + NavIC S 4-ch CRPA front end

Array and interface layout (antenna side, top view)

PART	DWG	REV	SHEET
CRL-LS-CRPA4	CRL-LS-CRPA4-DWG	A	2 / 6
DATE	UNITS	PROJ.	TOL.
06-11-2026	mm	1st angle	ISO 2768-mK



NOTES

1. All SMD parts on the bottom (L4). Reflow SAC305, type 4 paste, 0.10-0.12 mm stencil.
2. B7504 / B8889 MSL3: bake per J-STD-033 if exposed > 168 h. TA1442A peak 260 C, ESD sensitive.
3. THT after reflow: 8 x SMA Amphenol 132134, J5 Micro-Fit (selective / hand solder).
4. No flux residue on SAW lids. 100 % AOI; X-ray of the DFN-8, DL1721 and SAW ground pads.
5. Keep the exposed shield-frame lands free of solder, flux and coating (gasket seat).
6. Reference designators: CH n blocks repeat with sequential rotation; see BOM for values.
7. Parts list: CRL-LS-CRPA4_BOM.xlsx (40 lines, 313 placements); CPL: _CPL_bottom.csv.

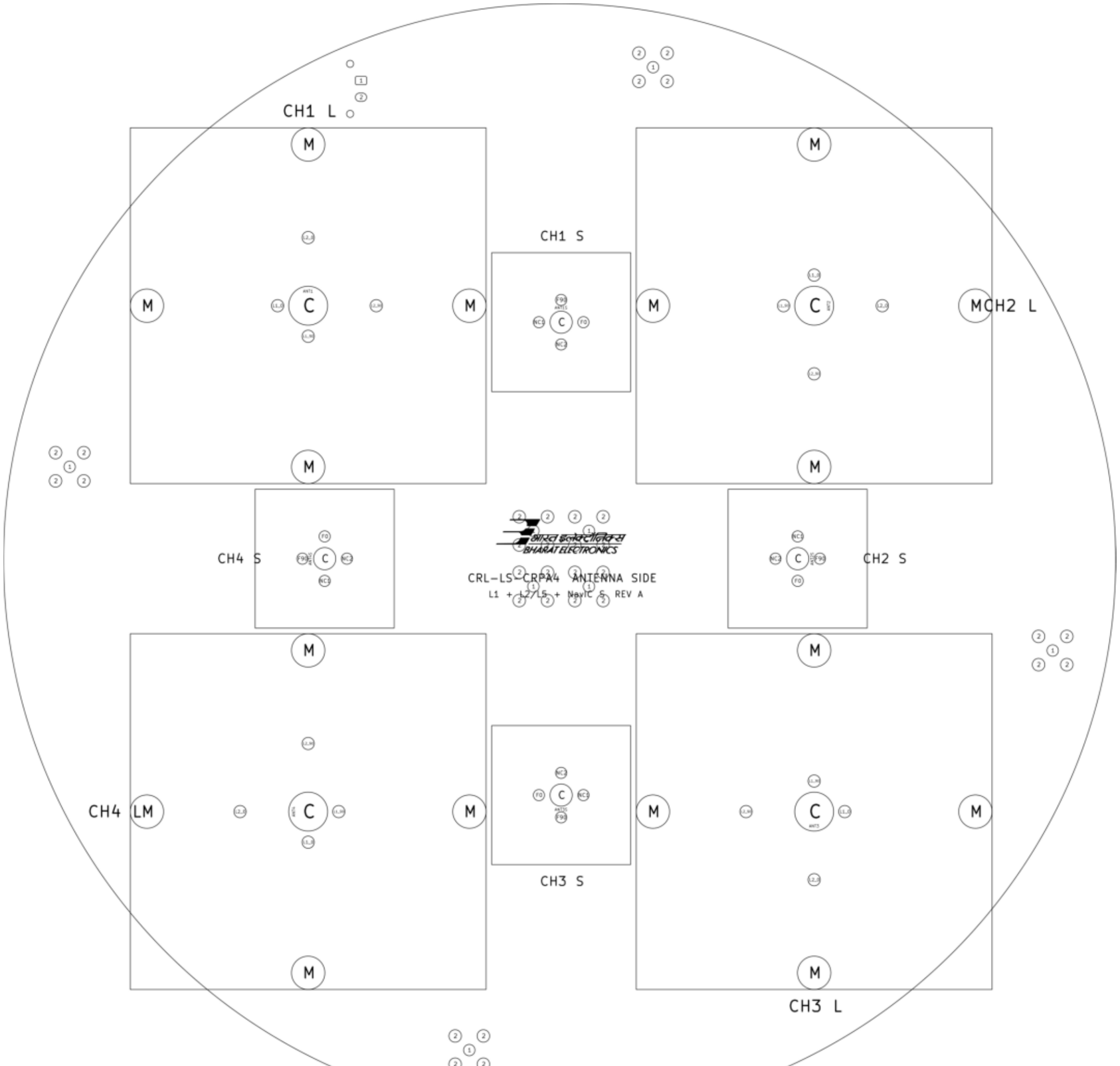
BEL - CRL

ANTENNA / RFM LAB

L1 + L2/L5 + NavIC S 4-ch CRPA front end

Assembly drawing, bottom (component side, viewed from below)

PART	DWG	REV	SHEET
CRL-LS-CRPA4 PCA	CRL-LS-CRPA4-DWG	A	3 / 6
DATE	UNITS	PROJ.	TOL.
06-11-2026	mm	1st angle	ISO 2768-mK



NOTES

1. Fit after bottom-side assembly and test.
2. 4 x GPDF6010.A L1/L2 patches: M3 through the D3.6 holes into the back cover bosses.
3. 4 x CRL-LS-SP01 NavIC S patches: pins through D0.77, solder from the bottom; centre D2.2 GND.
4. Solder the L feed pins (D1.35) from the bottom; trim pins <= 1.0 mm proud.
5. Top copper L1 is the solid antenna ground plane: no damage, no coating under the patches.
6. BEL logo and title on the top legend.

BEL - CRL

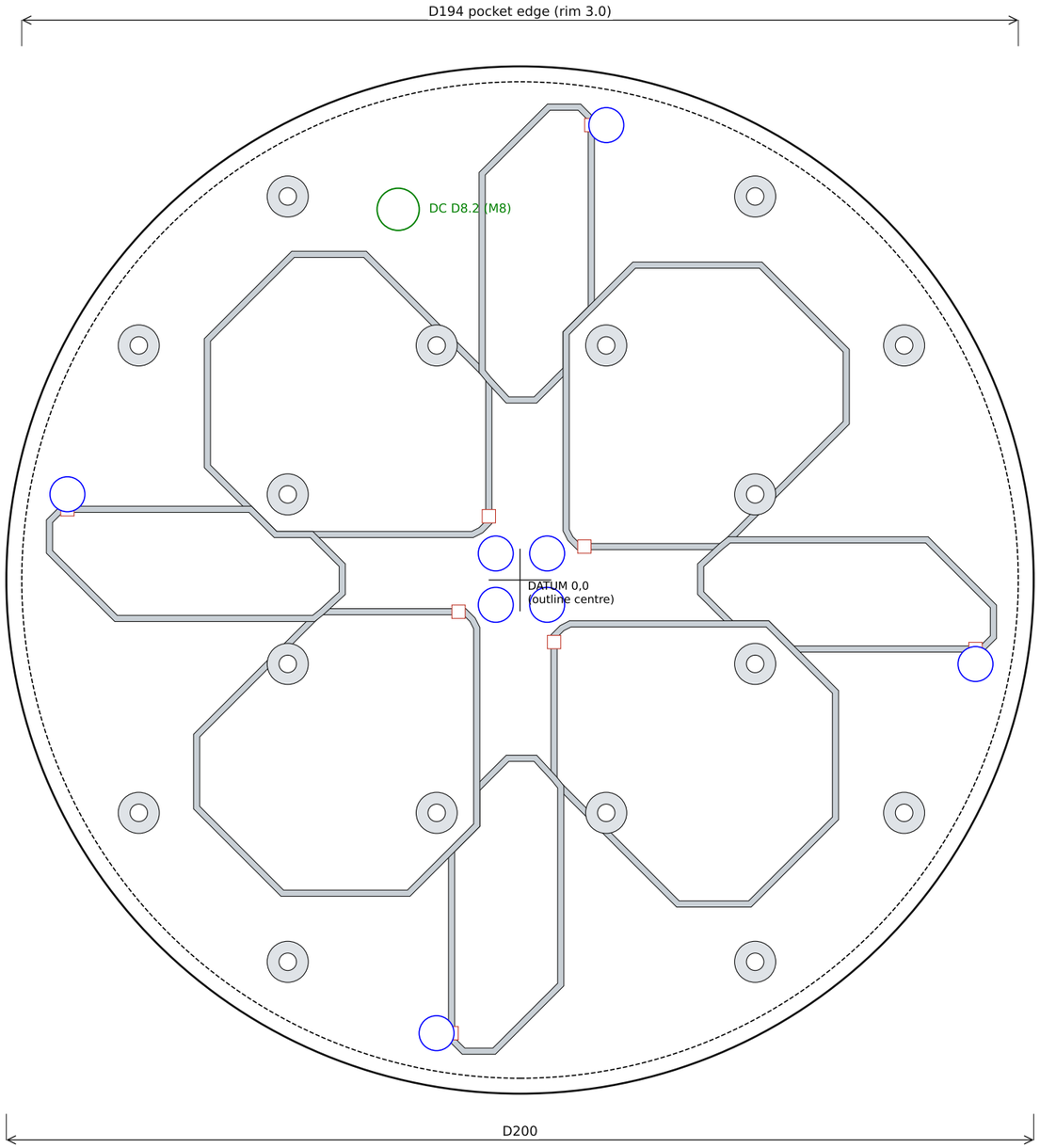
ANTENNA / RFM LAB

L1 + L2/L5 + NavIC S 4-ch CRPA front end

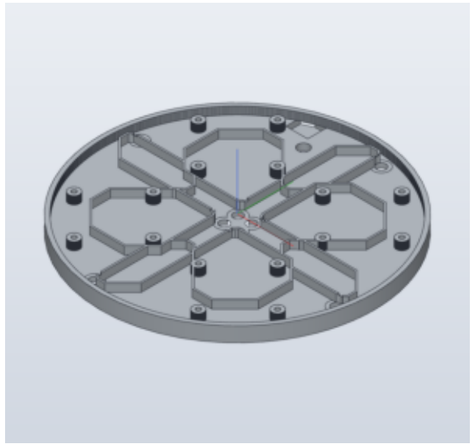
Assembly drawing, top (antenna side)

PART	DWG	REV	SHEET
CRL-LS-CRPA4 PCA	CRL-LS-CRPA4-DWG	A	4 / 6
DATE	UNITS	PROJ.	TOL.
06-11-2026	mm	1st angle	ISO 2768-mK

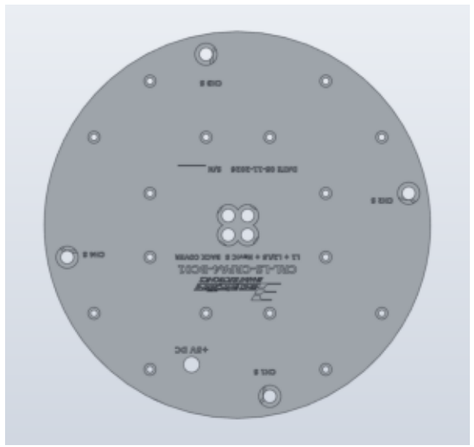
VIEW FROM THE BOARD SIDE (inside)



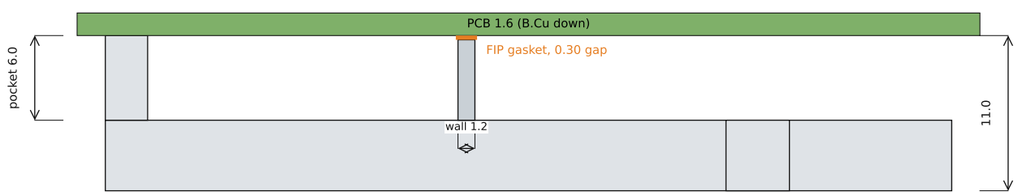
ISO, inside



outside face



SECTION (schematic, not to scale in X)



FEATURES

FEATURE	SIZE	QTY
M3 clearance + c'bore	D3.4 thru, D6.2 x 3.4 outside	16
SMA opening	D6.8 thru, D11.5 x 5.0 outside	8
DC receptacle	D8.2 thru (M8 panel)	1
RF tunnel under wall	2.6 x 1.0 mm	8
Boss at M3	D8.0 to the wall top	16

NOTES

1. Material Al 6061-T6. Chromate conversion MIL-DTL-5541 Type II Class 3 (conductive).
2. Mass 464 g (STEP). STEP: 06_step/CRL-LS-CRPA4-BC01.step, same frame as the PCB STEP.
3. Wall tops = shield-frame lands of the PCB (04_fab/_shield.json); flatness 0.05 over each wall.
4. Form-in-place conductive gasket CHOFORM 5575 [ASSUMED]: 0.8 bead, 0.30 compressed.
5. 16 x ISO 4762 M3 x 12 A2 (cover, board, L patch), 0.9 N m [ASSUMED], cross pattern.
6. Engraving 0.3 deep on the outside face: BEL logo, P/N, date, S/N, channel labels, +5V DC.
7. Break edges 0.2. ISO 2768-mK. Hole positions +-0.05 to datum 0,0.
8. DC receptacle and M2.5 flange holes per the chosen part [ASSUMED type].

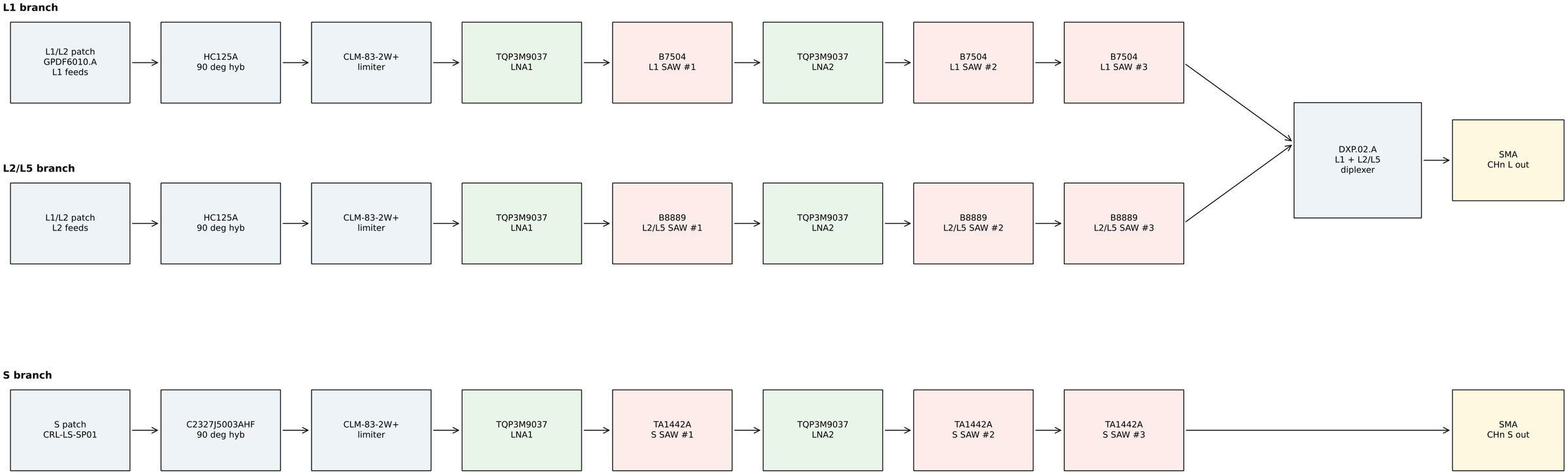
BEL - CRL

ANTENNA / RFM LAB

L1 + L2/L5 + NavIC S 4-ch CRPA front end

Back cover BC01 (L + S), machined

PART	DWG	REV	SHEET
CRL-LS-CRPA4-BC01	CRL-LS-CRPA4-DWG	A	5 / 6
DATE	UNITS	PROJ.	TOL.
06-11-2026	mm	1st angle	ISO 2768-mK



+5 V (J5, F1, Q1 reverse protection, D1 TVS) -> star bias on In1 to all 24 LNAs (18 nH / 5.6 nH chokes, 100 pF + 1 uF)

RF SUMMARY (antenna feed -> SMA, 25 C typ; rf_budget.py)

CASE	GAIN dB	NF dB	IP1dB dBm	IIP3 dBm	SAW max dBm	P ant -> SAW2 max
GPS L1 C/A (1575.42 MHz)	37.1	1.01	-20.4	-7.9	15	-27.4
GPS L5 (1176.45 MHz)	40.8	0.79	-23.9	-11.4	5	-40.9
GPS L2C (1227.60 MHz)	38.3	0.83	-22.9	-10.4	5	-39.9
NavIC S (2492.028 MHz)	33.0	1.35	-15.2	-2.2	10	-26.1

POWER / THERMAL

Power: 1.68 A typ / 2.16 A max at 5 V = 8.4 W typ / 11.3 W max (24 x TQP3M9037).

LNA junction: 0.35 W x 53.4 C/W = +18.7 C over case; Tj 103.7 C at 85 C case (limit 190 C).

Rejection table: 05_docs/CRL-LS-CRPA4_rejection.csv; full report: 05_docs/CRL-LS-CRPA4_design_report.pdf.

BEL - CRL

ANTENNA / RFM LAB

L1 + L2/L5 + NavIC S 4-ch CRPA front end

Block diagram (one channel, x4) and RF summary

PART	DWG	REV	SHEET
CRL-LS-CRPA4	CRL-LS-CRPA4-DWG	A	6 / 6

DATE	UNITS	PROJ.	TOL.
06-11-2026	mm	1st angle	ISO 2768-mK